Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The solubility of a solid in a liquid is significantly affected by temperature changes.
Solute + Solvent $\rightleftharpoons$ Solution.
The system being in a dynamic equilibrium must follow Le-chatelier's principle. Considering the Le-chatelier's principle which of the following is correct?

(A) $\Delta H_{sol} > 0$; solubility $\uparrow$; temperature $\downarrow$

(B) $\Delta H_{sol} < 0$; solubility $\downarrow$; temperature $\uparrow$

(C) $\Delta H_{sol} > 0$; solubility $\downarrow$; temperature $\uparrow$

(D) $\Delta H_{sol} < 0$; solubility $\uparrow$; temperature $\uparrow$

Q.2

Which of the following modes of expressing concentration is
independent of temperature?

(A) Molarity

(B) Molality

(C) Formality

(D) Normality

Q.3

How many grams of concentrated nitric acid solution should
be used to prepare $250\ mL$ of $2.0 M HNO_3$? The concentrated acid is $70\% HNO_3$

(A) $90.0\ g$ conc. HNO_3

(B) $70.0\ g$ conc. HNO_3

(C) $54.0\ g$ conc. HNO_3

(D) $45.0\ g$ conc. HNO_3

Q.4

Mole fraction of the solute in a 1.00 molal aqueous solution
is:

(A) 0.1770

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(B) 0.0177

(C) 0.0344

(D) 1.7700

Q.5

$25.3\ \mathrm{g}$ of sodium carbonate, $\mathrm{Na_2CO_3}$ is dissolved in enough water to make $250\ \mathrm{mL}$ of solution. If sodium carbonate dissociates completely, molar concentration of sodium ions, $\mathrm{Na^+}$ and carbonate ions, $\mathrm{CO_3^{2-}}$ respectively are (Molar mass of $\mathrm{Na_2CO_3} = 106\ \mathrm{g\ mol^{-1}}$)

(A) $0.955\mathrm{M}$ and $1.910\mathrm{M}$

(B) $1.910\mathrm{M}$ and $0.955\mathrm{M}$

(C) $1.90\mathrm{M}$ and $1.910\mathrm{M}$

(D) $0.477\mathrm{M}$ and $0.477\mathrm{M}$

Q.6

Which of the following statements, regarding the mole fraction (x) of a component in solution, is correct?

(A) $0 \leq \mathrm{x} \leq 1$

(B) $x \leq 1$

(C) x is always non-negative

(D) $-2 \leq x \leq 2$

Q.7

A solution is prepared by dissolving $10\ \mathrm{gNaOH}$ in $1250\ \mathrm{mL}$ of a solvent of density $0.8\ \mathrm{g/mL}$. The molality of the solution in $\mathrm{molkg^{-1}}$ is

(A) 0.25

(B) 0.2

(C) 0.008

(D) 0.0064

Q.8

Two solutions of a substance (non electrolyte) are mixed in the following manner. $480\ \mathrm{mL}$ of $1.5\mathrm{M}$ first solution $+520\ \mathrm{mL}$ of $1.2\mathrm{M}$ second solution. What is the molarity of the final mixture?

(A) $2.70\mathrm{M}$

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(B) $1.344\mathrm{M}$

(C) $1.50\mathrm{M}$

(D) $1.20\mathrm{M}$

Q.9

The mole fraction of water in 20% aqueous solution (By weight) of $\mathrm{H_2O_2}$ is

(A) $\frac{77}{68}$

(B) $\frac{68}{77}$

(C) $\frac{20}{80}$

(D) $\frac{80}{20}$

Q.10

Persons are medically considered to have lead poisoning if they have a concentration of greater than 10 micrograms of lead per decilitre of blood. Concentration in parts per billion is

(A) 1000

(B) 100

(C) 10

(D) 1

Q.11

A binary liquid solution is prepared by mixing n -heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solution?

(A) The solution is non-ideal, showing - ve deviation from Raoult's Law.

(B) The solution is non-ideal, showing $+$ ve deviation from Raoult's Law.

(C) n -heptane shows $+$ ve deviation while ethanol shows - ve deviation from Raoult's Law.

(D) The solution formed is an ideal solution.

Q.12

Which among the following will show maximum osmotic pressure?

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(A) $1 \mathrm{MNaCl}$

(B) $1 \mathrm{MMgCl}_2$

(C) $1 \mathrm{M}\left(\mathrm{NH}_4\right)_3 \mathrm{PO}_4$

(D) $1 \mathrm{MNa}_2 \mathrm{SO}_4$

Q.13

Formation of a solution from two components can be considered as
(i) Pure solvent $\rightarrow$ separated solvent molecules, $\Delta \mathrm{H}_1$
(ii) Pure solute $\rightarrow$ separated solute molecules, $\Delta \mathrm{H}_2$
(iii) Separated solvent and solute molecules $\rightarrow$ Solution, $\Delta \mathrm{H}_3$ Solution so formed will be ideal if

(A) $\Delta \mathrm{H}_{\text{soln}} = \Delta \mathrm{H}_3 - \Delta \mathrm{H}_1 - \Delta \mathrm{H}_2$

(B) $\Delta \mathrm{H}_{\text{soln}} = \Delta \mathrm{H}_1 + \Delta \mathrm{H}_2 + \Delta \mathrm{H}_3$

(C) $\Delta \mathrm{H}_{\text{soln}} = \Delta \mathrm{H}_1 + \Delta \mathrm{H}_2 - \Delta \mathrm{H}_3$

(D) $\Delta \mathrm{H}_{\text{soln}} = \Delta \mathrm{H}_1 - \Delta \mathrm{H}_2 - \Delta \mathrm{H}_3$

Q.14

Which of the following liquid pairs shows a positive deviation from Raoult's law?

(A) Water - Nitric acid

(B) Benzene - Methanol

(C) Water - Hydrochloric acid

(D) Acetone- Chloroform

Q.15

A solution of acetone in ethanol

(A) shows a positive deviation from Raoult's law

(B) behaves like a non ideal solution

(C) obeys Raoult's law

(D) shows a negative deviation from Raoult's law

Q.16

An aqueous solution is 1.00 molal in KI. Which change will cause the vapour pressure of the solution to increase?

(A) Addition of NaCl

(B) Addition of $\mathrm{Na}_2 \mathrm{SO}_4$

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(C) Addition of 1.00 molal KI

(D) Addition of water

Q.17

P_A and P_B are the vapour pressure of pure liquid components, A and B, respectively of an ideal binary solution. If X_A represents the mole fraction of component A, the total pressure of the solution will be.

(A) $P_A + X_A (P_B - P_A)$

(B) $P_A + X_A (P_A - P_B)$

(C) $P_B + X_A (P_B - P_A)$

(D) $P_B + X_A (P_A - P_B)$

Q.18

Vapour pressure of chloroform $(CHCl_3)$ and dichloromethane (CH_2Cl_2) at $25^\circ C$ are $200\ mmHg$ and $41.5\ mm$ Hg respectively. Vapour pressure of the solution obtained by mixing $25.5\ g$ of $CHCl_3$ and $40\ g$ of CH_2Cl_2 at the same temperature will be: (Molecular mass of $CHCl_3 = 119.5u$ and molecular mass of $CH_2Cl_2 = 85u$).

(A) $90.92\ mmHg$

(B) $115.0\ mmHg$

(C) $147.9\ mmHg$

(D) $285.5\ mmHg$

Q.19

Which condition is not satisfied by an ideal solution?

(A) $\triangle_{mix} V = 0$

(B) $\triangle_{mix} S = 0$

(C) Obeyance to raoult's law

(D) $\triangle_{mix} H = 0$

Q.20

The Van't Hoff factor of solutes A, B and C in aqueous solutions are 0.8, 1.6 and 1.2 respectively. The freezing point of equimolar solutions follow the order :

(A) $A > B > C$

(B) $A > C > B$

(C) $B > A > C$

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(D) $B > C > A$

Answers & Solutions

Q.1 Answer:

$\Delta H_{sol} < 0$; solubility $\downarrow$; temperature $\uparrow$

Solution:

According to Le-chateliers principle, for an exothermic reaction $(\Delta H < 0)$ increase in temperature decreases the solubility.

Q.2 Answer:

Molality

Solution:

The molality involves weights of the solute and the solvent. Since weight does not change with the temperature, therefore molality does not depend upon the temperature.

Q.3 Answer:

45.0 g conc. HNO_3

Solution:

$$\textbf{Molarity } (M) = \frac{wt \times 1000}{\text{mol. wt.} \times \text{vol (mL)}}$$

$$2 = \frac{\text{wt.}}{63} \times \frac{1000}{250}$$

$$wt = \frac{63}{2} \text{ g}$$

$$\textbf{wt. of } 70\% \textbf{ acid} = \frac{100}{70} \times 31.5 = 45 \text{ g}$$

Q.4 Answer:

0.0177

Solution:

1 molal solution means 1 mole of solute dissolved in 1000gm solvent.

$$\therefore n_{solute} = 1 \quad w_{solvent} = 1000 \text{ g}$$

$$\therefore n_{solvent} = \frac{1000}{18} = 55.56$$

$$X_{solute} = \frac{1}{1+55.56} = 0.0177$$

Q.5 Answer:

$1.910M$ and $0.955M$

Solution:

Concentration of

$$Na_2CO_3 = \frac{25.3}{106} \times \frac{1000}{250} = 0.955M$$

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$\left[\mathrm{Na}^{+}\right]=2 \times 0.955=1.91 \mathrm{M}$
$\left[\mathrm{CO}_3^{2-}\right]=0.955 \mathrm{M}$

Q.6 Answer:

x is always non-negative

Solution:

Mole fraction of any component A
$\mathrm{x}=\frac{\text{No. of moles of A}}{\text{Total No. of moles}}$
As total no. of moles $>$ **no. of moles of** A
thus x **can never be equal to one or zero.**

Q.7 Answer:

0.25

Solution:

Given $\mathrm{w}=10 \mathrm{~g}, \mathrm{Mol.}$ **mass** $=40 \mathrm{~g}$
Weight of solvent $=1250 \times 0.8 \mathrm{~g}=10000 \mathrm{~g}$
$=1 \mathrm{~kg}$
$\therefore$ **molality** $=\frac{10}{40 \times 1}=0.25 \mathrm{~m}$

Q.8 Answer:

$1.344 \mathrm{M}$

Solution:

From the molarity equation.
$\mathrm{M}_1 \mathrm{~V}_1+\mathrm{M}_2 \mathrm{~V}_2=\mathrm{MV}$
$\mathrm{M}=\frac{\mathrm{M}_1 \mathrm{~V}_1+\mathrm{M}_2 \mathrm{~V}_2}{\mathrm{~V}}$ **where** $\mathrm{V}=\mathrm{V}_1+\mathrm{V}_2$
$\mathrm{M}=\frac{480 \times 1.5+520 \times 1.2}{480+520}=1.344 \mathrm{M}$

Q.9 Answer:

$\frac{68}{77}$

Solution:

Mole fraction of $\mathrm{H}_2\mathrm{O}=\frac{\frac{80}{18}}{\frac{80}{18}+\frac{20}{34}}=\frac{68}{77}$

Q.10 Answer:

100

Solution:

$10 \mu \mathrm{g}$ **per decilitre**
$\Rightarrow 10 \times 10^{-6} \mathrm{~g}$ **in** $100 \mathrm{~mL}$

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$\therefore 10^9$ **parts** $(1$ **billion** $)$ **has** $= \frac{10 \times 10^{-6}}{100} \times 10^9 g = 100 g$

Q.11 Answer:

The solution is non-ideal, showing $+$ ve deviation from Raoult's Law.

Solution:

For this solution intermolecular interactions between n -heptane and ethanol are weaker than n -heptane- n heptane \& ethanol-ethanol interactions hence the solution of n -heptane and ethanol is non-ideal and shows positive deviation from Raoult's law.

Q.12 Answer:

$1\mathrm{M}\ (\mathrm{NH_4})_3\,\mathrm{PO_4}$

Solution:

Vant Hoff factor $i = 4$ in case of $(\mathrm{NH_4})_3\,\mathrm{PO_4}$,
$(\mathrm{NH_4})_3\,\mathrm{PO_4} \rightleftharpoons 3\mathrm{NH_4^+} + \mathrm{PO_4^{3-}}$

Q.13 Answer:

$\Delta \mathrm{H_{soln}} = \Delta \mathrm{H_1} + \Delta \mathrm{H_2} + \Delta \mathrm{H_3}$

Solution:

For an ideal solution, $\Delta \mathrm{H_{mixing}} = 0\ \Delta \mathrm{H} = \Delta \mathrm{H_1} + \Delta \mathrm{H_2} + \Delta \mathrm{H_3}$
(According to Hess's law)
i.e., for ideal solutions there is no change in magnitude of the attractive forces in the two components present.

Q.14 Answer:

Benzene - Methanol

Solution:

Positive deviations are shown by such solutions in which solvent-solvent and solute-solute interactions are stronger than the solute-solvent interactions. In such solution, the interactions among molecules becomes weaker. Therefore their escaping tendency increases which results in the increase in their partial vapour pressures. In pure methanol there exists intermolecular H-bonding. On adding benzene, its molecules come between methanol molecules there by breaking H-bonds which weaken intermolecular forces. This results in increase in vapour pressure.

Q.15 Answer:

shows a positive deviation from Raoult's law

Solution:

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A solution of acetone in ethanol shows positive deviation from Raoult's law. It is because ethanol molecules are strongly hydrogen bonded. When acetone is added, these molecules break the hydrogen bonds and ethanol becomes more volatile. Therefore its vapour pressure is increased.

Q.16 Answer:

Addition of water

Solution:

When the aqueous solution of one molal KI is diluted with water, concentration decreases, therefore the vapour pressure of the resulting solution increases.

Q.17 Answer:

$\mathrm{P_B + X_A\,(P_A - P_B)}$

Solution:

$$\begin{aligned}
&\mathrm{P = P_A X_A + P_B X_B} \\
&\mathrm{= P_A X_A + P_B\,(1 - X_A)} \\
&\Rightarrow \quad \mathrm{P_A X_A + P_B - P_B X_A} \\
&\Rightarrow \quad \mathrm{P_B + X_A\,(P_A - P_B)}
\end{aligned}$$

Q.18 Answer:

$90.92\ \mathrm{mmHg}$

Solution:

$$\begin{aligned}
&\mathrm{n_{CHCl_3}} = \frac{25.5}{119.5} = 0.213 \\
&\mathrm{n_{CH_2Cl_2}} = \frac{40}{85} = 0.47 \\
&\mathrm{P_T = P_A^o X_A + P_B^o X_B} \\
&= 200 \times \frac{0.213}{0.683} + 41.5 \times \frac{0.47}{0.683} \\
&= 62.37 + 28.55 = 90.92
\end{aligned}$$

Q.19 Answer:

$\triangle_{\mathrm{mix}}\ \mathrm{S} = 0$

Solution:

For an ideal solution,

Volume change $\triangle\ \mathrm{V} = 0$ of mixing should be zero.

Heat change $\triangle\ \mathrm{H} = 0$ on mixing should be zero.

Obey Raoult's law every range of concentration.

Q.20 Answer:

$\mathrm{A > C > B}$

Solution:

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